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(1) Research Program/Project/Study Title: Pre-Service Science Teacher's Scientific Self-Efficacy and Cognitive Anxiety in Relation to their Science Engagement

Proponent Name:	Fe Mae L. Laborte
Academic Rank:	Asso. Prof. V
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader/s:	
Academic Rank:	

(2) Category (please check)

- ☒ Social Science/Education /Humanities & other related fields
- ☐ Natural/Biological Sciences
- ☐ Food Research
- ☐ Developmental/ Computer/Engineering/ Architecture
- ☐ Extension

(3) Research Center

[Check the concerned center]

☒ SSITRDC
Social Science and
Industrial Technology
Research Development
Center

☐ LRDC
Livestock Research
Development Center

☐ FFRDC
Fisheries and Food
Research Development
Center

☐ CSRDC
Crop Science Research
Development Center

(4) Site of implementation

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Municipality/City: Pontevedra	
Province: Capiz	
Region: VI	
(5) Classification Research: ☒ Basic ☐ Applied Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity Education	(8) Discipline/Field of Specialization Teaching and Learning
(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.) Science education provides students the opportunity to gain better knowledge of how and why things function. It teaches several valuable skills, finding solutions to a problem and fast paced discovery and development. Nevertheless, teaching and learning science becomes a challenge to both teachers and students. Understanding science (as content, inquiry and process skills) is a challenging task for students, as it involves a construction process, which is complex and iterative where it takes time and effort (Hadzigeorgiou Y and Schulz RM, 2019). According to the 2022 PISA results, the Philippines ranked poorly in the 2022 Programme for International Student Assessment (PISA) among 81 participating countries. The Philippines ranked third from the bottom in science with an average score of 356, sixth from the bottom in mathematics with an average score of 355, and sixth from the bottom in reading with an average score of 347 (Montemayor, 2023). Moreover, World Health Organization (WHO) reported that in the world, one in six people aged 10–19 years experiencing physical, emotional and social changes, these includes poverty, abuse, or violence. Hence, adolescents are vulnerable to mental health problems. Furthermore, globally it is estimated that 1 in 7 (14%) 10–19 year-olds experience mental health conditions, yet these remain largely unrecognized and untreated. Thus, adolescents with mental health conditions are particularly vulnerable to social exclusion, discrimination, stigma (affecting readiness to seek help), educational difficulties, risk-taking behaviours, physical ill-health and human rights violations (WHO, 2021). Mental health is one of the problem that affects students' education which reflects to their performance in every subjects. Mathematics and science where most students develop anxiety, often they lose focus and confidence to immerse their selves in engaging to the said subjects. This study will try to find the relationship between self-efficacy and cognitive anxiety in relation to their science engagement.	
(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project	

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objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned. .)

N/A

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

This study will be conducted to find significant relationship between scientific self-efficacy and cognitive anxiety in relation to Science Engagement. More specifically, this study seeks to find answers to the following questions:

1. What is the extent of the respondents' scientific self-efficacy as a whole and when they are grouped according to their demographic profile?
2. What is the extent of the respondents' cognitive anxiety as a whole and when they are grouped according to their demographic profile?
3. What is the extent of the respondents' science engagement as a whole and when they are grouped according to their demographic profile?
4. Is there a significant relationship between the respondents' scientific self-efficacy and their science engagement?
5. Is there a significant relationship between the respondents' cognitive anxiety and their science engagement?

(11) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

This study is anchored to the following theories:

1. Self-efficacy theory emphasizes the importance of the individual and the individual's perceptions of his/her personal capabilities as key determinants of successful outcomes. Self-efficacy theory, and the broader social cognitive theory in which self-efficacy is encompassed, therefore clearly endorses a democratic ideal that suggests that all individuals are competent and capable of being successful, provided they have the opportunities and self-efficacy necessary to pursue their goals (Gallagher, 2012). As mentioned by Gil, 2004 individuals are willing to make will depend on the level of self-efficacy for the task. If they perceive themselves as having high self-efficacy for the task, they will probably be willing to work hard when they perform alone or alongside coactors (social facilitation). Individuals in this situation feel responsible and self-confident about their performance. In contrast, individuals who believe that they have low self-efficacy for the task are less likely to work hard, especially when their performance is being evaluated (social loafing).

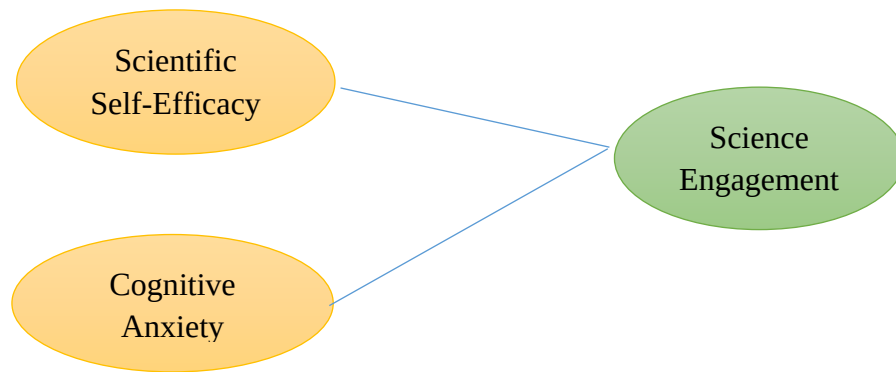
2. Learning engagement is a positive, substantial, stable, and lasting mental state related to learning. Learning engagement is not only an important predictor of academic achievement, but also an effective observation index to measure learning quality and predict the development and growth of students (Yang, et al 2021).

3. Social cognitive theory by Albert Bandura describes the influence of individual experiences, the actions of others, and environmental factors on individual health behaviors. SCT provides opportunities for social support through instilling expectations, self-efficacy, and using observational learning and other reinforcements to achieve behavior change.

4. Cognitive theory posits that an extensive cognitive system exists that has a taxonomical structure, varying from surface level thoughts (products), to mechanisms that operate on information (processes), and deep structures (schemas). Cognitive structures drive the manner in which information is processed which, in turn, manifest as cognitive products (Rnic & Dazois, 2017)

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Conceptual Framework



(12) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Review of Literature

Pre-Service Teachers and Philippine Science Performance

In the Philippine context, all teacher education curriculum in all higher education institutions included practicum teaching as one of the requirements for the degree of Bachelor of Secondary Education (BSEd). It prepares student-teachers to become qualified teachers in the future. As cited by Yesilcinar & Aykan, 2022, each education system aims to train individuals who can collaborate and communicate appropriately as well as think critically and creatively which is one of the 21st-century skills. This points out the vital role of teacher training programmes because they are supposed to prepare pre-service teachers with the required knowledge and skills. The results of exams such as PISA and TIMSS bring the education systems of the countries in the first place to the fore. Researchers consider the lesson study model as the reason why these countries are at the top in the list of PISA and TIMSS (Aykan & Dursun, 2021). Although the collaborative and communication-based environment offered by the lesson study model can contribute to both critical and creative thinking skills of pre-service teachers.

According to the 2022 PISA results, the Philippines ranked third from the bottom in science with an average score of 356, sixth from the bottom in mathematics with an average score of 355, and sixth from the bottom in reading with an average score of 347 (Montemayor, 2023).

It is also noted that 94 percent of 15-year-old Filipino students spoke a language other than English at home, which was the highest proportion among all the countries that were assessed except for Lebanon. Another alarming finding from the assessment was that 65 percent of Filipino participants reported being bullied at least a few times a month, compared to an average of just 23 percent across OECD countries. It reported that private school students in the country significantly outperformed public school students across all three categories: reading (390 vs 328), mathematics (395 vs 343) and science (399 vs 347). It also noted that the number of Philippine respondents was smaller than most other participating countries, which it said could be due to many 15-year-olds having dropped out of school.

Science Engagement

One of the challenges faced by science educators was on how to engage students in science and helping them develop an understanding of its ideas. Since, Science is one of the subjects to be taught and learned. Understanding science (as content, inquiry and process skills) is a challenging task

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for students, as it involves a construction process, which is complex and iterative—not a linear one—and which also takes time and effort. In one of the study made by constructivist-oriented research in the 1980s and 1990s showed, is the possibility for students to *construct* not only a conceptual framework that lacks the coherence of true scientific knowledge, but to equally construct alternative ideas that are different from the canonical scientific ones (Hadzigeorgiou Y and Schulz RM, 2019). Moreover, other consequences were discovered like in the construction process, it is influenced by several interrelated factors, such as students' prior conceptions and views on the nature of science their interest and motivation, the classroom culture, the opportunities they have for social interaction, dialogue, and argumentation, the generation of representations (for the use of modeling and analogies), and also their opportunities for cognitive dissonance and conceptual change, as well as for applying new knowledge to new contexts (Kalman 2017).

The research teams have taken one of the nine aspects of science engagement (science-related activities outside of school) and equated that aspect alone to science engagement. OECD as cited by Jack et. Al (2014) has described science engagement as including behavioral, cognitive and emotional factors are identified weak relationships between science self-concept, science self-efficacy, science enjoyment and science interest and what they termed “science engagement” (actually science-related activities). Notably, this is an incremental portion of what other authors refer to as science engagement; in this case, a behavioral measure, outside of the context of formal schooling, and not including either cognitive or emotional measures. Thus, its use as an outcome measure as a proxy for science engagement seems questionable at best. In some cases, all or nearly all of the other aspects of science engagement were shown to be correlated to science-related activities outside of school (yet the relationships of these measures with science achievement were left untested). These correlations lend some support to OECD's inclusion of science-related activities as part of science engagement.

Mental Health

Because anxiety doesn't discriminate. Anyone can experience anxiety, and any number of things — or combination of things — can cause or trigger it. And childhood anxiety can affect all areas of a child's life. It can have negative effects on social, physical, emotional, and learning performance. In 2015, research showed that 6.8% of Australia children suffered from an anxiety disorder. And all signs point to these numbers growing. The following are ways on how anxiety affects learning; (1) Anxiety may make a child uncomfortable in the school environment. (2) Anxiety affects working memory - our ability to hold information in our minds for short periods, in order to do something with it. (3) Anxiety makes thought processing less efficient. (4) Anxiety can be hard to detect. (5) Anxiety can lead to avoidance — of school or homework.

Anxiety is not a life sentence. It can come and go, and it can be treated. But early intervention is vital to stopping anxiety from getting worse. And this means giving kids the tools and strategies they need to deal with anxiety and stress – now and in the future. The following are the things that can parents do to help their children : (1) Develop a solid, consistent and productive family routine. (2) Talk to your kids about school (and everything), often. (3) Practice mindfulness.(4) On that note, limit screen time.(5)Make sure they are getting enough sleep. (6) Be engaged with their schooling. (7) Praise effort, not results. (8) Seek help if they — or you — need it.

Cognitive Anxiety

Anxieties, in some form or another, have been believed to exist ever since the first human. In many ways, this type of physiological reaction to stimuli was an evolutionary advantage for our ancestors. Anxiety and fear responses kept early humans alive and were critical in our development by allowing us to recognize dangers and sense possible threats. Those traits still exist in our genes and are still being expressed in our behaviors. In fact, anxiety and fear are closely related in their mechanisms. The difference between anxiety and fear is that anxiety is a more diffuse and generalized emotional response, while fear is a response to a particular circumstance (Horwitz, 2013).

Science anxiety is just one such example and follows close on the heels of math and statistics anxiety, test anxiety, and general school anxiety. In fact, many of these educational anxieties stem from the feeling of being out of control of a situation (Kurbanoglu, 2014). The things that set these anxieties apart however are the symptoms under which anxiety is felt. Science anxiety as the disgust or fear of science concepts, scientists, and scientific activities. This broad set of symptoms is often narrowed by defining the field of science and the specific circumstances of the feelings.

Some blame the No Child Left Behind (NCLB) Act for causing some of these problems As schools struggled to meet the expectations of the Act, they took time away from non-required subjects,

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like science, to focus solely on those needed to meet the standards, such as reading and math scores. As reported by elementary school teachers, science was deemed a secondary subject at the initial startup of NCLB, and less classroom time was dedicated to teaching science. This, in turn, has led to a lack of preparation of students to enter higher levels of science education and, in many cases, potential anxiety over the material (Milner et al 2012).

It is the job of science educators to try to motivate students and overcome the fear of the material, but with students entering science courses already pre-disposed to fear due to poor preparation by prior educators, social stigma and cultural indoctrination, science education then becomes an uphill struggle. Hinds, suggests that science educators need to “seize the moment” in a science course and address needs and fears of the students at the start. While this is an excellent idea, it still does not deal with the established fear most students have of the material. Without understanding how the students deal with the established anxieties over the subject material and knowing how it may or may not affect their learning process, educators have difficulty determining how to help students.

Scientific Self-Efficacy

Science self-efficacy was described as a student’s “confidence in performing science-related tasks” (OECD, 2009). Based on that concept, found that mastery experiences were most responsible for the formation of science self-efficacy. Chen and Pajares (2010) determined that a student’s belief in her incremental abilities would enhance science self-efficacy; Utito (2014) related this aspect to a student’s perception of his capacity to put into play acquired knowledge and skills when confronted with new science-based tasks. Thus, a consensus definition for science self-efficacy could be stated as a student’s confidence in performing challenging new science-related tasks.

Science self-concept is a student’s belief that she can easily learn and understand science concepts. Steinmayr, Dinger, and Spinath (2012) argued that the dimensions of science self-concept should be considered as intrinsic task value, importance, and utility. Some authors identified science self-concept primarily as an expectation for successful achievement in science. Jansen, Schroeders and Lüdtke (2014) considered biology, chemistry and physics self-concepts separately; however, they simply used the PISA 2006 items for science self-concept and replaced “science” with “biology” or “chemistry” or “physics” in their items. The definition for science self-concept could thus be extended as follows: science self-concept is a student’s belief that he can learn, understand, and perform at a high level in science disciplines.

According to Hamlyn et-al (2020) students typically accessed science content through reading about it online, TV or streaming, books, newspapers or magazines and students who had visited a science-related attraction or activity such as a science museum or festival, suggests an important role for these channels in widening access to and engagement in science.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

This research study adopted the descriptive-correlational research design measures a relationship between two variables without the researcher controlling either of them. To identify variables that have some sort of relationship do the extent that a change in one creates some change in the other. This study also includes a description, analysis and interpretation of the data gathered.

This study will be conducted to randomly Pre-service Science Teachers of Capiz State University for SY 2023-2024.

The study will utilize an adopted and modiefied questionnaire about self efficacy, cognitive anxiety and science engagement. The questionnaire will be subjected to reliability test.

The following statistical tools were employed in the description and analysis of the data gathered.

Frequency Count. This will be used to determine the number of cases that belong to a certain category of variable.

Percentage score. This will be used to express the ratio of the score obtained in an item/component to the total score of the item/component

Mean. This will be used to reflect in a single value the overall performance of the respondents in a particular item, area, or entire component of the research instruments.

Standard Deviation. This statistical measure will be used to describe the variability or dispersions of test scores from the mean.

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Inferential Statistics. This will be used to test hypothesis, such as correlations between different responses or patterns in the whole dataset. The test is done at 5 percent level of significance.

Regression Analysis. This tool will be used to examine the relationship between two or more variables of interest.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart. Below is an example. This can be modified base on your desired Work plan.)

Research Objectives	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Field work/observation												
Analysis of data gathered from observation												
Writing of Summary, conclusion & Recommendation												
Editing												
Paper Preseantation												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

None

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

N/A

(16) Targets Beneficiaries (Who the clienteles are and what are the expected outcome/effects of the use of the projects output)

The result of this study will serve as a baseline information that benefits the following:

Parents- the study will provide platform for parents to help their children to be proficient and have interest in engaging to science subject.

Students- the study will give them the idea on how they will cope up their science engagement putting barrier to anxieties towards the science subject. The data gathered will also have students' improve both academic and personal growth.

Schools- this study will improve the school to develop amd foster new ways of enhancing knowledge, skills and attitudes regarding to the students' anxiety to science subject.

Future Researcher- this study will provide them a useful reference information in supporting their study in improving science literacy of the students in the country.

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
Fe Mae L. Laborte	Study Leader	Science

Percent time devoted to the Program/ Project/ Study:

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

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SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		P 2,000.00
- Meal allowance during data gathering		1,500.00
- Professional fees (laboratory analysis and other related fees)		
- Miscellaneous (for incidental expenses)		P 1, 500.00
- Internet		
- Printing		P 2, 000.00
- Paper presentation/publication		P 70, 000.00
C. CAPITAL OUTLAY		
Total		77, 000.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

1. Aykan & Dursun. (2021). Investigating Lesson Study Model within the Scope of Professional Development in Terms of Pre-Service Teachers. *International Online Journal of Educational Sciences* 13(5):1388-1408 DOI:[10.15345/iojes.2021.05.007](https://doi.org/10.15345/iojes.2021.05.007)
2. Hadzigeorgiou Y and Schulz RM (2019). Engaging Students in Science: The Potential Role of "Narrative Thinking" and "Romantic Understanding". *Front. Educ.* 4:38. doi: 10.3389/feduc.2019.00038
3. Hamlyn et-al (2020). Young people's views on science education. Kantar, Public Division. 215 Euston Road, London NW1 2BE, UK). E-7214.3/03-2020/RK
4. Horwitz, A. V. (2013). The sociological study of mental illness: A critique and synthesis of four perspectives. In C. S. Aneshensel, J. C. Phelan, & A. Bierman (Eds.), *Handbook of the sociology of mental health* (2nd ed., pp. 95–112). Springer Science + Business Media. https://doi.org/10.1007/978-94-007-4276-5_6
5. Jack, RE., et. al (2014). Dynamic facial expressions of emotion transmit an evolving hierarchy of signals over time. 24(2):187-192. doi: [10.1016/j.cub.2013.11.064](https://doi.org/10.1016/j.cub.2013.11.064)
6. Kalman, C. S. (2017). *Successful Science and Engineering Teaching: Theoretical and Learning Perspectives, 2nd Edn.* New York, NY: Springer.
7. Montemayor, MT., (2023) CHED to address PH students' low int'l assessment ranking. <https://www.pna.gov.ph>
8. Uitto (2014)
9. WHO, (2021). Mental health of Adolescents. <https://www.who.int/news-room/fact-sheets/detail/adolescent-mental-health>
10. Yang, C., et al (2021). College students' stress and health in the COVID-19 pandemic: the role of academic workload, separation from school, and fears of contagion. *PLoS ONE* 16, e0246676. doi: 10.1371/journal.pone.0246676
11. Yesilcinar, S., & Aykan, A. (2022). Lesson Study and 21st-Century Skills: Pre-Service Teachers Reason, Produce and Share. *Participatory Educational Research*, 9(3), 315-329.

(20) Capsule Curriculum Vitae (One-page brief C.V. only)

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FE MAE L. LABORTE

I. Personal Information

Date of Birth: November 5, 1978
 Place of Birth: Bailan, Pontevedra, Capiz
 Civil Status: Married
 Address: Blk11 Lot 3, Villa de San Lorenzo Ruiz 1,
 Cagay, Roxas City
 Academic Rank: Asso. Prof V
 Specialization: Science
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II. Educational Background

Doctoral Degree:	Ph. D.	Major: Science Ed.(Science)
School:	WVSU	
Master's Degree:	MAT	Major: Gen. Science
School:	Capiz State University Pontevedra Capiz	
Bachelor's Degree:	BSBS	Major: Bio Science
School:	WVSU	

III. Designations/ Current Position: QA and Internal Quality Auditor